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High-fidelity embedded gas sensing: a unified deep learning framework for simultaneous identification and quantification

Xuanque Nguyen¹ , Ngoc Viet Nguyen² , Chien Nguyen Viet² , Van Su Luong² ,
Quang Vuong Pham² , Viet Thong Le² , Van Hieu Nguyen² and Minhhuu Le^{2,*}

¹ Faculty of Information System, Phenikaa School of Computing, Phenikaa University, Hanoi, Vietnam

² Faculty of Electrical and Electronic Engineering, Phenikaa School of Engineering, Phenikaa University, Hanoi, Vietnam

* Author to whom any correspondence should be addressed.

E-mail: huy.leminh@phenikaa-uni.edu.vn

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Abstract

Electronic nose systems are a transformative technology, enabling low-cost, real-time gas analysis for critical applications in environmental monitoring, industrial safety, and medical diagnostics. However, their practical performance is fundamentally limited by the computational challenge of analyzing time-series data from their sensor arrays. Conventional machine learning (ML) pipelines rely on lossy feature engineering and employ separate, inefficient models for gas classification and concentration regression, a methodology that fundamentally limits predictive accuracy. This paper presents a unified multitask deep learning model that solves these problems through three key innovations. First, the model learns features directly from the complete, raw sensor time-series using a convolutional neural network, avoiding information loss from pre-processing. Second, a novel attention mechanism links the classification and regression tasks, allowing the model to use the identified gas type to improve its concentration estimate. Third, a custom loss function was developed to ensure high accuracy across a wide range of gas concentrations, a critical challenge in chemical sensing. Using 5-fold cross-validation, the model achieved 100% accuracy in gas classification and a mean absolute percentage error (MAPE) of 6.61% for concentration regression, results that significantly outperform classical ML benchmarks. We also demonstrated the model's efficiency by deploying it on 10 different STM32 microcontrollers, confirming its suitability for real-time embedded applications. This work provides a robust framework for developing high-performance, AI-based chemical sensors and offers a foundational methodology for future analysis of complex gas mixtures. The source code and data are provided in this GitHub repo: https://github.com/quenx-rd/gas_identifier_public.

1. Introduction

Accurate gas detection and quantification are essential in a variety of domains, including environmental safety, industrial monitoring, and healthcare. Toxic gases such as carbon monoxide (CO), nitrogen dioxide (NO₂), and volatile organic compounds pose significant risks to human health and ecosystems [1–5]. The detection of specific gases in exhaled breath has enabled early diagnosis of respiratory diseases [2–5]. Conventional methods for gas analysis, such as gas chromatography and mass spectrometry, offer high accuracy but are often bulky, expensive, and unsuitable for real-time or field deployment [6–8]. To address these limitations, researchers have developed electronic nose (e-nose) systems that emulate human olfactory processing by integrating sensor arrays with AI-based pattern recognition algorithms [9–12]. These systems typically rely on a set of gas sensors with partially overlapping sensitivities, allowing them to detect and classify a wide range of gas species.

Metal–oxide–semiconductor (MOS) sensors are widely used in e-nose platforms due to their low cost, compact form factor, high sensitivity, and ease of integration [13–15]. Commercial MOS sensors

such as the MiCS-4514 and MiCS-6814 from SGX Sensortech are popular choices for both academic and industrial gas sensing applications [16]. MOS sensors operate by modulating their electrical resistance in response to gas exposure. Their performance can be enhanced by applying temperature modulation, which varies the sensor's operating temperature to generate dynamic, time-dependent responses commonly referred to as thermal fingerprints [17–19]. These fingerprints contain rich information about gas-sensor interactions and can improve the selectivity of a sensor, especially when combined with machine learning (ML) techniques [20–22].

A wide range of traditional ML algorithms has been applied to analyze sensor data and distinguish between gas types. Common techniques include principal component analysis (PCA) [19–21], linear discriminant analysis (LDA) [19, 22–24], support vector machines [25, 26], k-nearest neighbors (kNN) [27], multilayer perceptron (MLP) [28], quadratic discriminant analysis (QDA) [29], Gaussian Naive Bayes (GNB) [30], random forest (RF) [31], decision tree (DT) [32], and gradient boosting (GB) [25]. These approaches have shown strong potential in classifying gases based on static or dynamic sensor response patterns. A recent study [33] constructed an e-nose with eight MOS sensing elements (from four MiCS-4514 chips) and evaluated multiple ML models for classification and regression. Those approaches are marked by a critical deficiency: a reliance on reductive feature engineering. In this process, the raw time-series data is condensed into a small set of handcrafted, static features—such as peak response, rise time, or integral values derived from dimensionality reduction techniques like PCA. This is fundamentally a lossy compression, discarding subtle yet discriminative information contained in the shape and evolution of the sensor response and making the model's performance highly dependent on the selection of these features. In addition, those pre-processing methods are time consuming (i.e. PCA) and might not be possible to deploy on low-resource microcontrollers (MCUs) in the e-nose system. Beyond the inherent limitations of this feature engineering process, the modeling strategy itself presents a second fundamental deficiency: the disjointed treatment of gas identification and quantification. The common practice is to train separate models for the classification task (identifying the gas) and the regression task (estimating its concentration). This separation is not only inefficient, leading to increased training and deployment overhead, but it is also informationally suboptimal. The regression model operates in isolation, unaware of the classification output, despite the fact that the physical relationship between sensor response and concentration is inherently gas-specific.

Recent advancements (2023–2025) have shifted toward deep learning to overcome these deficiencies. 1D-convolutional neural network (CNN) architectures have achieved 100% accuracy in coffee variety classification [34], while lightweight variants like LwCNN enable on-site perfume identification on edge modules [35]. To capture temporal dependencies, hybrid CNN-BiLSTM models have been deployed for gas leakage detection [36], and federated learning frameworks have been introduced to preserve data privacy [37]. Furthermore, the integration of attention mechanisms, such as in TFA-CNN, allows for enhanced feature mining in both time and frequency domains for simultaneous gas recognition and concentration prediction [38]. Lightweight Transformer-based models like ConvTran have also demonstrated state-of-the-art accuracy in capturing global dependencies on embedded e-nose platforms. TinyML has further enabled real-time diagnostics directly on MCUs for applications like diabetes detection via breath analysis [39].

Despite these advancements, several critical research gaps persist in the existing literature (see detailed comparison table in table 1). First, there is a scarcity of unified multitask architectures capable of performing both gas classification and concentration regression simultaneously, as these tasks are often handled by discrete, independent systems. Second, there is a near absence of works integrating multitask deep learning (MT-DL) with actual MCU deployment for real-time, low-power edge analytics, which is essential for decentralizing data processing. Finally, the use of rigorous validation procedures, such as leave-one-concentration-out, to analyze model generalization on completely unseen and non-overlapping data remains rare in current research.

To address these challenges, this paper introduces a unified, end-to-end MT-DL framework that directly learns from raw thermal fingerprints. Unlike prior work, we eliminate handcrafted features and disjointed pipelines, replacing them with a single deep neural network capable of simultaneous gas classification and quantification. Our architecture leverages 1D convolutional (Conv1D) networks for automatic feature extraction and employs a dual-branch multitask design. A novel attention mechanism allows the classification output to dynamically condition the regression task, ensuring synergistic learning across both objectives. Furthermore, we propose a concentration-adaptive loss function that maintains high prediction fidelity across wide dynamic ranges, with particular robustness in low-concentration regimes. Finally, we validate the framework's embedded feasibility by deploying it on 10 STM32 MCUs, demonstrating its practicality for real-world, resource-constrained edge AI applications. The contributions that realize this methodology are fourfold:

Table 1. Comparative analysis of recent e-nose studies (2023–2025). The bolded entries indicate data identical to those of the proposed work.

Study (year)	Sensor type	No. of gases	Task type	Data size	Best performance	Embedded support
Stefanone (2025) [40]	MOS (BME688)	4 classes (4 odors)	Classification	4750 samples	96.42% macro-F1	Yes (Pi 3)
Chen (2025) [41]	Acoustic signal	12 classes	Classification	684 sets	97.81% accuracy	No
Iswanto (2024) [34]	MOS (MQ series)	2 classes (2 odors)	Classification	80 samples	100% accuracy	Yes (Arduino)
El Barkani (2024) [42]	MOS (MQ series)	4 classes (3 mixtures + air)	Classification	6400 samples	91.73% accuracy	Yes (Arduino Nano)
Gudiño-Ochoa (2024) [39]	MOS (MQ series)	2 classes (healthy + diabetes)	Classification	44 individuals	95.00% accuracy	Yes (Arduino Nano)
Jiang (2024) [38]	MOS (In2O3/SnO2)	5 classes (5 gases)	Multitask	250 samples	100% accuracy, 0.99 R^2	No
Sharma (2024) [37]	MOS (MQ series)	4 classes (3 mixtures + air)	Classification	6400 samples	99.70% accuracy	No
Zhao (2023) [43]	FET sensor array	2 classes (2 mixtures)	Multitask	350 samples	98.4% accuracy, 0.99 R^2	No
Srivastava (2023) [44]	MOS (thick-film)	4 classes (4 gases)	Multitask	58 samples	100% accuracy, MSE	No
Attallah (2023) [36]	MOX	5 classes (4 mixtures + air)	Classification	6400 samples	99.25% accuracy	No
Cao (2023) [35]	MOS (Figaro, SGX)	6 classes (6 mixtures)	Classification	300 samples	98.35% accuracy	Yes (Jetson Nano)
Proposed Work	MOS (MiCS-4514)	9 classes (8 gases + air)	Multitask	594 samples	100% accuracy, 6.61% MAPE	Yes (10 STM32 MCUs)

Employing a deep convolutional network to autonomously learn hierarchical feature representations from the complete, raw thermal fingerprints, preserving their full temporal integrity and maximizing information extraction.

Introducing a novel attention mechanism that functionally links the classification and regression tasks, enabling the model to explicitly condition its concentration predictions on the identified analyte for improved accuracy.

Designing a custom loss function that resolves the critical challenge of wide dynamic ranges by ensuring high-fidelity quantification across all concentration scales, particularly in low-concentration regimes.

Validating the model's computational efficiency by deploying it on a diverse suite of 10 STM32 MCUs, confirming its feasibility for real-world, resource-constrained edge AI applications.

2. Data preparation

2.1. Experimental setup

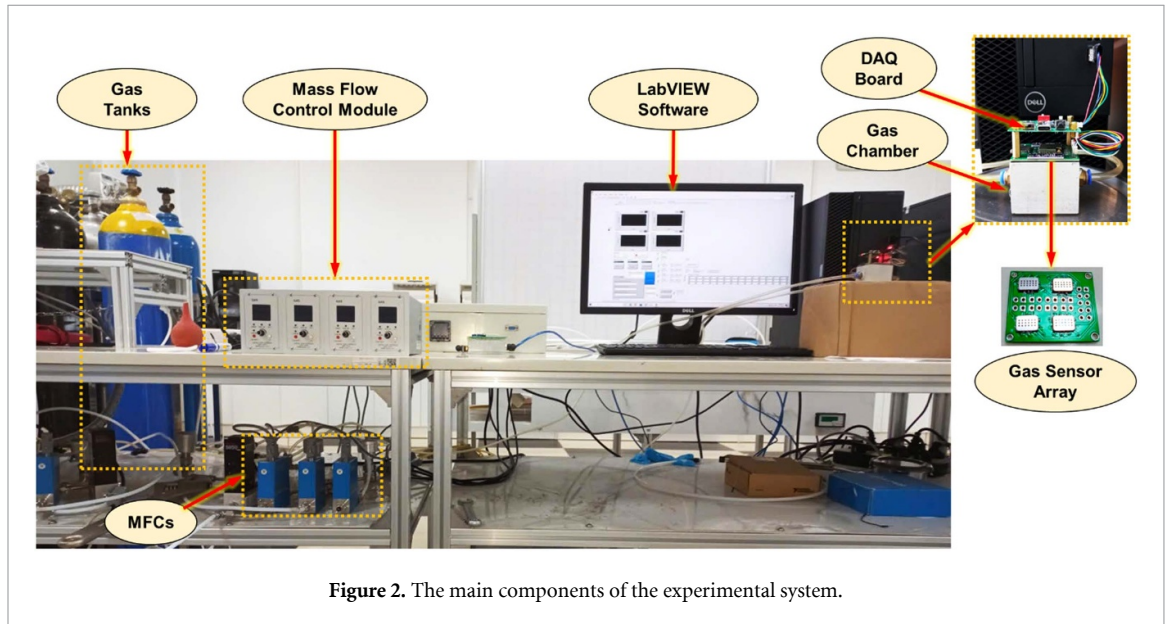
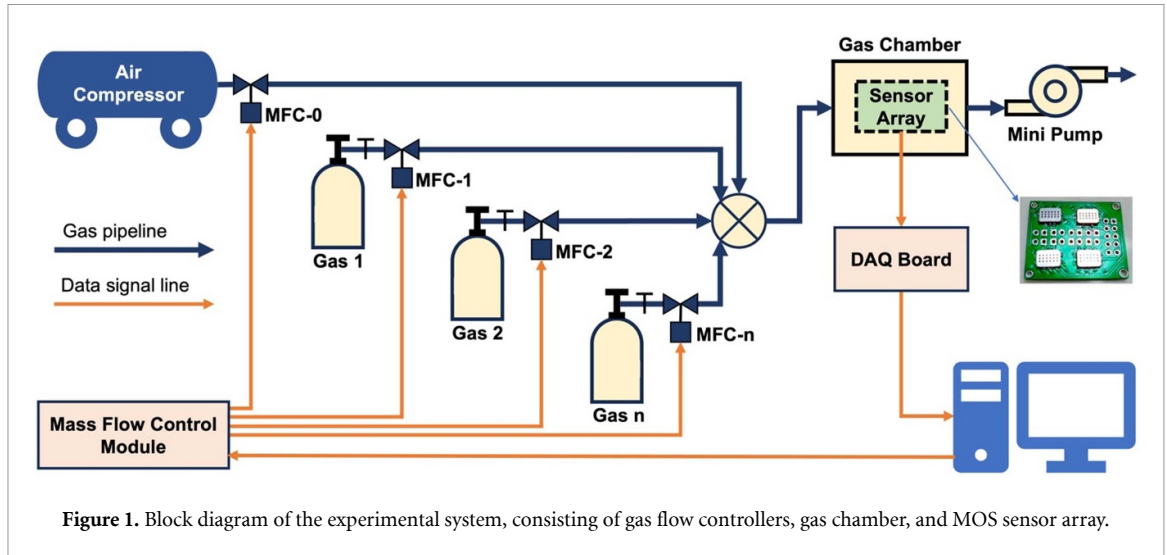
Figures 1 and 2 show the block diagram a photograph of the actual experimental setup of the gases measurement system, respectively. The system integrates a sealed sensing chamber within a computer-controlled continuous flow system, enabling precise delivery of target gases while maintaining a constant total flow rate [33]. The system consists of pressurized gas cylinders connected to mass flow controllers (MFCs) and a mass flow control module, which regulates the precise mixing and delivery of target gases. The conditioned flow is directed into the sealed gas chamber containing the MOS sensor array. Data acquisition is managed by a dedicated data acquisition board, while a computer equipped with a LabVIEW software controls the flow sequence, records sensor responses, and synchronizes the measurement cycles. Together, these components enable stable, repeatable, and automated operation of the gas sensing experiments.

2.2. Sensor array characterization and inter-sensor correlation

Gas delivery is regulated by multiple MFCs with maximum capacities of 500, 100, and 10 ml min⁻¹ ($\pm 1\%$ accuracy, Kojima Instruments Inc., Kyoto, Japan). Each MFC is connected to a pressurized cylinder containing either clean dry air or a target gas diluted in air. Clean background air is produced using a compressor, filtration system, and dry air generator, conditioned to 10% relative humidity at 25 ± 1 °C. The sensor array consists of four MiCS-4514 chips (SGX Sensortech), providing eight independent MOS sensing elements (denoted S1–S8). Each element outputs an electrical resistance that varies with gas exposure. The baseline resistance in clean air is denoted R_a , and the resistance under target gas exposure is denoted R_g . To reduce the effect of sensor variability and baseline drift, responses are normalized as R_g/R_a . Unless otherwise specified, all figures report normalized values.

Target gases—CO, H₂, NH₃, C₂H₅OH, CH₃COCH₃, H₂S, SO₂, and NO₂—were individually introduced and diluted with carrier air to achieve six concentration levels. A fixed total flow rate of 400 ml min⁻¹ was maintained throughout all experiments. A total of eight experiments were conducted, one for each gas, where each gas was measured continuously across its six concentration levels, as shown in figure 3. Each target gas was measured once per concentration level, providing a single raw sample prior to data augmentation. Each concentration measurement cycle consisted of a 90 s exposure phase followed by a 180 s recovery phase in clean air with sample rating 1 Hz. The transient response is characterized by a rise behavior during the initial part of the 90 s exposure as the gas concentration in the chamber stabilizes, while the decay behavior begins immediately after 90 s when clean air is introduced for recovery. Prior to experiments, the sensor array was preheated and stabilized in clean air to ensure repeatability.

To characterize the relationships within the sensor array, we computed the Pearson correlation matrix for the complete dataset (figure 4). The analysis reveals a highly structured relationship, identifying distinct clusters of correlated sensors rather than a set of independent channels. For instance, a tightly coupled cluster comprises sensors S2, S3, S6, and S8, with pairwise correlations as high as 0.98 (e.g. between S2 and S3), indicating significant response redundancy. A second cluster is formed by S4, S5, and S7, which also exhibit strong inter-correlations (e.g. 0.90 between S4 and S5). In stark contrast, sensor S1 shows weak correlation with most other elements, indicating that it provides a unique and non-redundant information channel. This high degree of collinearity presents a challenge for traditional linear models but also provides an opportunity for a more sophisticated approach to exploit this structure for robustness. This complex correlation landscape underscores the necessity of a data-driven model like our proposed deep learning framework, which is capable of autonomously learning to navigate these



inter-sensor dependencies, effectively leveraging both the redundant and unique information to build a highly discriminative representation of the gas fingerprint.

To further characterize the array, we analyzed the linear correlation between each sensor's response and the concentration of each target gas, we computed the point-biserial correlation coefficient, a special case of the Pearson correlation used when one variable is continuous (the sensor signal) and the other is dichotomous (the presence or absence of a specific gas). The calculation was performed for each sensor-gas pair as follows. For a given gas G , a binary indicator vector was created, with a value of 1 for all samples corresponding to that gas and 0 otherwise. The point-biserial correlation coefficient ($r_{S,G}$) was then computed between the signal vector of a given sensor S and the binary indicator vector for gas G across all N samples in the dataset. It is mathematically expressed as equation (1),

$$r_{S,G} = \frac{\sum_{i=1}^N (S_i - \bar{S}) (G_i - \bar{G})}{\sqrt{\left(\sum_{i=1}^N (S_i - \bar{S})^2\right) \left(\sum_{i=1}^N (G_i - \bar{G})^2\right)}} \quad (1)$$

where S_i is the sensor value at sample i , and G_i is the binary indicator (1 for gas G , 0 otherwise). This metric effectively measures how much a sensor's signal tends to increase or decrease when a specific gas is present, relative to the sensor's average signal across all conditions.

The resulting coefficients are visualized in the heatmap in figure 5. The results highlight the fundamental challenge of MOS sensor selectivity. No single sensor exhibits a strong, exclusive correlation with any single analyte, confirming the cross-sensitive nature of the array. However, the matrix does reveal

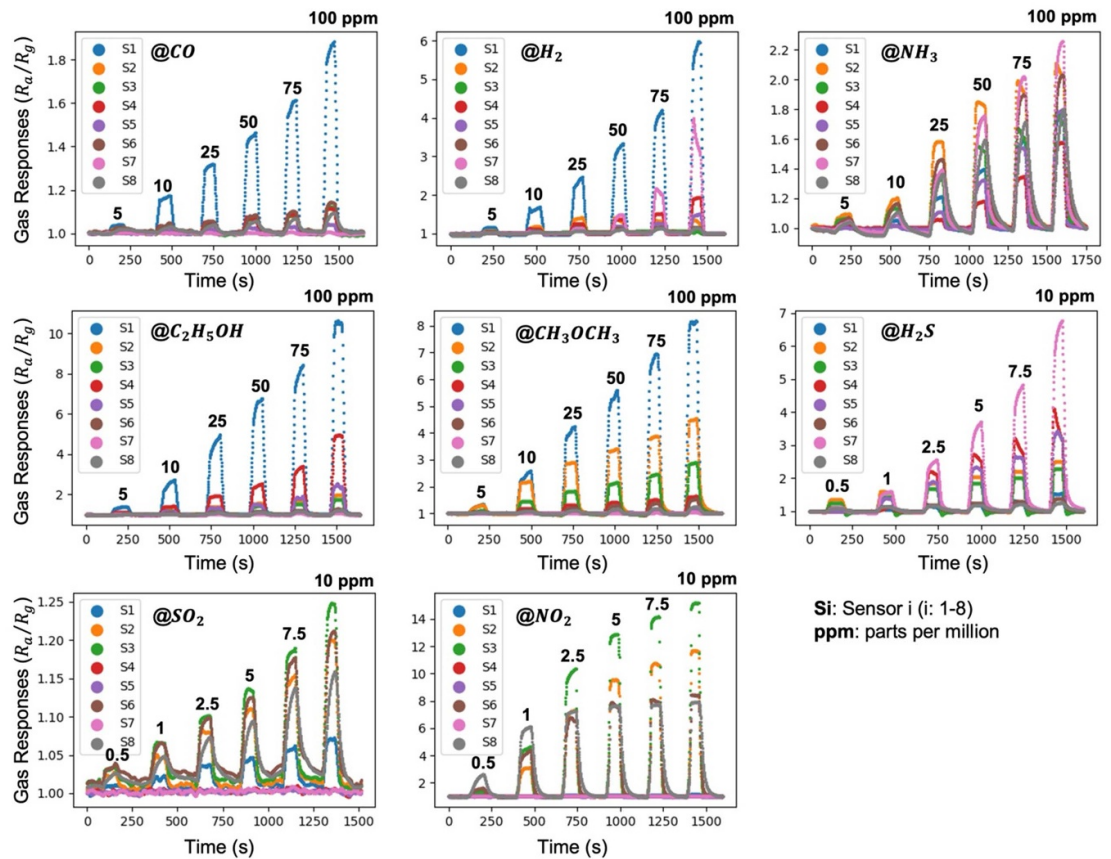


Figure 3. Continuous responses of the eight MOS sensing elements (S1–S8) to selected target gases. For each gas, six concentration levels were measured sequentially. The Y-axis shows normalized resistance values (R_g/R_a).

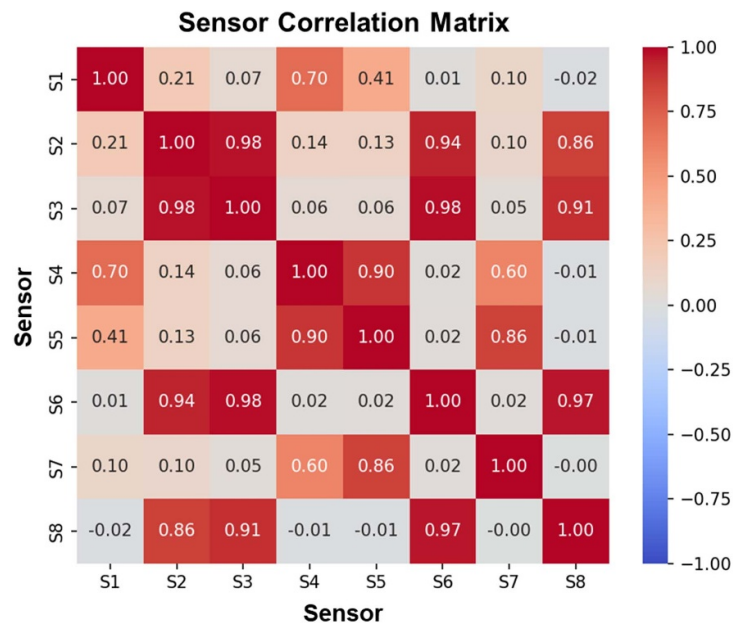
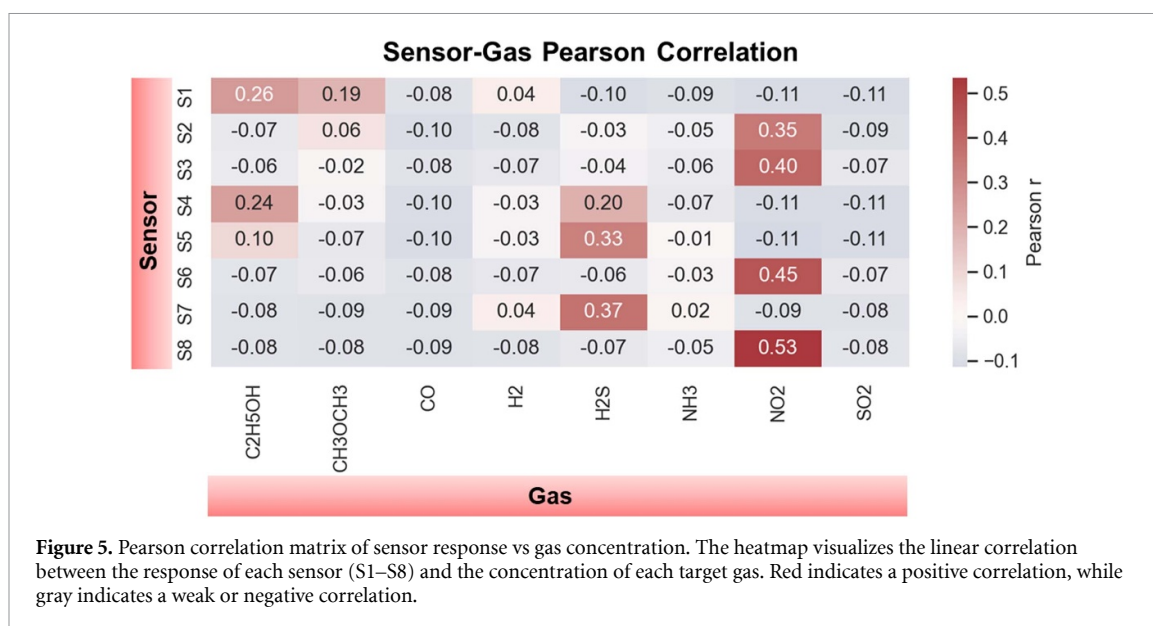


Figure 4. Pearson correlation matrix of the eight MOS sensor elements (S1–S8). The heatmap visualizes the pairwise correlation coefficients, revealing significant redundancy and distinct sensor groupings within the array. Dark red indicates a strong positive correlation (approaching +1.0), while lighter colors indicate weak or no correlation.

specialized sensitivities; for instance, NO_2 elicits the strongest positive correlations across the array, particularly with sensor S8 ($r = 0.53$) and S6 ($r = 0.45$). Similarly, H_2S shows a moderate positive response from sensors S7 ($r = 0.37$) and S5 ($r = 0.33$). Conversely, the weak or negative correlations observed



for many other gas-sensor pairs (e.g. vs CO, H₂) strongly suggest that the underlying relationships are highly nonlinear. This analysis provides a clear justification for our modeling approach: the combination of low selectivity and high nonlinearity necessitates a sophisticated pattern recognition model. A deep learning framework is ideally suited to this task, as it can learn the complex, nonlinear decision boundaries from the collective response patterns of the entire array, a capability beyond that of simpler linear models.

2.3. Data augmentation through temporal segmentation

Deep neural networks require large, diverse datasets to learn generalizable features and avoid overfitting, a condition rarely met by the limited number of measurements typically collected in a laboratory setting [45, 46]. To address the data scarcity inherent in experimental gas sensing and to enhance the robustness of our deep learning model, we employed a sliding window data augmentation technique. We systematically expand the dataset by generating multiple, temporally shifted samples from each original measurement.

Each original measurement constitutes a time-series capturing the sensor array's dynamic response to a gas concentration. The core response, from initial rise to final decay, typically spans approximately 105 data points. So, we defined a slightly larger analysis window of 115 data points to capture the full signal shape while keeping the data collection time as short as possible, which was systematically translated across a broader temporal range with a fixed step size of 1 data point encompassing both the primary response and the adjacent baseline periods. This configuration results in an overlap rate of 99.13% (114 out of 115 points) between consecutive windows. This process segments each original measurement into overlapping time-series samples, effectively increasing the training dataset size by an order of magnitude. After augmentation, each gas concentration level contributes 11 samples, resulting in 66 samples per gas across all six concentrations. Figure 6 shows an example of the sliding window process.

This data augmentation technique serves a dual purpose. First, it introduces temporal translation invariance. By presenting the model with the gas signature at various positions within the window, it forces the model to learn the intrinsic shape and features of the response rather than its absolute temporal location. This is a critical feature for real-world deployment, where response timings may vary due to minor fluctuations in flow rate or environmental conditions. Second, and critically, this method allows for the systematic generation of samples corresponding to a 'clean air' or null class. By extracting windows exclusively from the baseline periods where no target analyte is present, we train the model to explicitly distinguish between the presence and absence of a target gas. This represents a significant methodological improvement over the baseline study [33], which did not include an air class, and is essential for preventing false positives in a practical e-nose system. The resulting augmented dataset provides the scale and diversity required to train a deep neural network that is both accurate and generalizable.

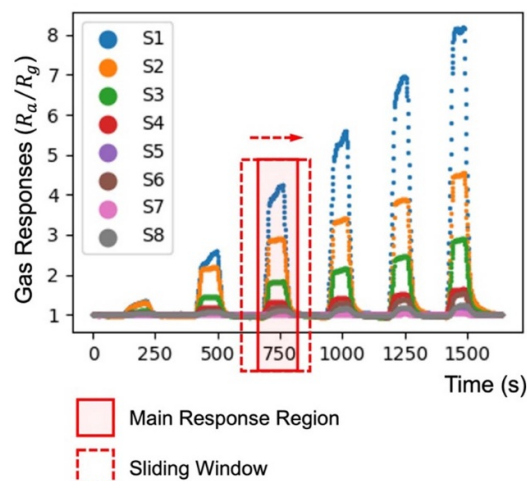


Figure 6. Example of sliding-window augmentation applied to sensor responses.

	Split 1:						Split 2:						Split 3:					
	C1	C2	C3	C4	C5	C6	C1	C2	C3	C4	C5	C6	C1	C2	C3	C4	C5	C6
CO	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
H ₂	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
NH ₃	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
C ₂ H ₅ OH	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
CH ₃ OCH ₃	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
H ₂ S	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
SO ₂	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
NO ₂	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11
Air	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11	11

For Training Set

Always For Training Set

For Validation/Test Set

Figure 7. Illustration of the cross-validation data splitting strategy. ‘C’ denotes concentration level. The test concentration for each gas rotates systematically across splits (e.g. C1, C2, and C3 of CO are held out for testing in splits 1, 2, and 3, respectively). The maximum concentration ($C_{\max}$) is permanently reserved for the training set to ensure the regression task remains an interpolation problem. Each cell represents 11 augmented samples.

2.4. Cross-validation

To rigorously assess model generalization and ensure the reliability of our performance metrics, we employed a 5-fold cross-validation scheme. Given the finite size of the experimental dataset, a robust cross-validation strategy is essential to obtain an unbiased estimate of the model’s performance on unseen data. A simple random split would be inadequate, as it would not guarantee that the model is tested on its ability to predict concentrations it has not been trained on. Therefore, we designed a specialized stratified, leave-one-concentration-out partitioning strategy. The augmented dataset was divided into five folds. In each iteration, one fold served as the held-out validation/test set, while the other four were used for training. The construction of these folds was highly structured to prevent informational leakage and to create a challenging and realistic evaluation scenario: 1) for each of the eight target gases, a single, unique concentration level was held out for testing in each split. For example, in split 1, the test set could include concentration level 1 of CO, concentration level 2 of H₂, and concentration level 3 of NH₃, while in split 2, a different combination of concentration levels from each gas was held out, and so on, for the first five concentration levels, as shown in figure 7; 2) this rotating hold-out strategy ensures that the model is always evaluated on its ability to interpolate—that is, to predict a concentration value it has not explicitly seen during training. This is a far more stringent and practical test of generalization than evaluating on a random subset of known concentrations.

A key constraint in our splitting logic is that the highest concentration level ($C_{\max}$) for each gas was always included in the training set for every fold. This is a deliberate design choice to frame the regression task as an interpolation problem—predicting values within a known operational range $(0, C_{\max}]$ —and not an extrapolation problem. This aligns with the intended real-world application of a calibrated e-nose system. This meticulous partitioning scheme guarantees that every concentration level is eventually used for testing, providing a comprehensive assessment across the dynamic range while rigorously preventing data leakage. The final performance metrics reported in this paper are aggregated from the predictions across all five distinct test folds, resulting in a single, robust estimate of the model's true generalization capability.

2.5. Data normalization

To ensure numerical stability and promote efficient model convergence, we applied distinct normalization schemes to the input sensor data and the target concentration labels. Proper normalization is critical for training deep neural networks, particularly when input features and output targets span different orders of magnitude, as is the case in this study. As shown in figure 8, normalization places all sensor responses on comparable scales. This improves training stability, accelerates convergence, and prevents any single sensor with a larger signal magnitude from dominating the feature representation, thereby forcing the model to learn consistent patterns from the entire array rather than being biased by arbitrary value ranges. For the regression task, each concentration label y is rescaled to the interval $(0, 1]$ by dividing by its maximum concentration $C_{\max}$, as expressed in equation (2). This guarantees a uniform target range across all gases. Without this scaling, since different gases have varying concentration ranges, the model would inherently prioritize analytes with higher absolute concentration values during the regression training, leading to significant performance degradation for gases measured at lower concentration scales. For the input sensor data, z-score normalization is applied so that each sensor feature has zero mean and unit variance, as expressed in equation (3–(5).

$$y_{\text{scaled}} = \frac{y}{C_{\max}} \quad (2)$$

$$\mu = \frac{1}{N} \sum_{i=1}^N X_i \quad (3)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2} \quad (4)$$

$$X_{\text{norm}} = \frac{X - \mu}{\sigma + \epsilon} \quad (5)$$

where X is the input, μ is the mean, σ is the standard deviation, and $\epsilon = 10^{-9}$ prevents division by zero. These statistics are computed only from the training set and then applied to both training and validation sets to avoid data leakage. The input data has shape $(N, 115, 8)$, where N is the number of samples, 115 is the number of time points per sample, and 8 is the number of sensors. Normalization is applied across both the sample and temporal dimensions, so each sensor channel is standardized according to its global distribution.

3. A unified framework for multitask gas classification and regression

To address the limitations of classical, disjointed modeling, we developed a unified, end-to-end MT-DL framework. The core of this framework is a single, cohesive neural network designed to learn directly from raw sensor time-series data and simultaneously perform two distinct but related tasks: identifying the gas analyte (classification) and estimating its concentration (regression). The architecture is specifically engineered to enable synergistic learning between these tasks, leading to improved performance and efficiency, notably by reducing prediction costs and latency through the consolidation of both tasks into a single forward pass.

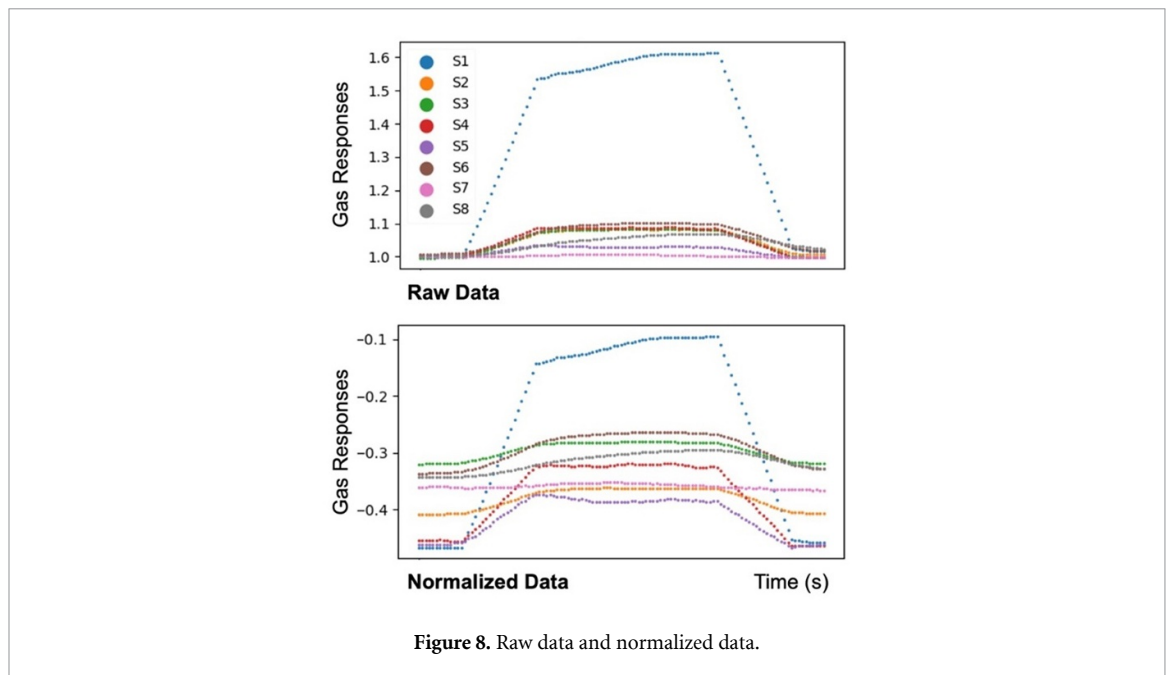


Figure 8. Raw data and normalized data.

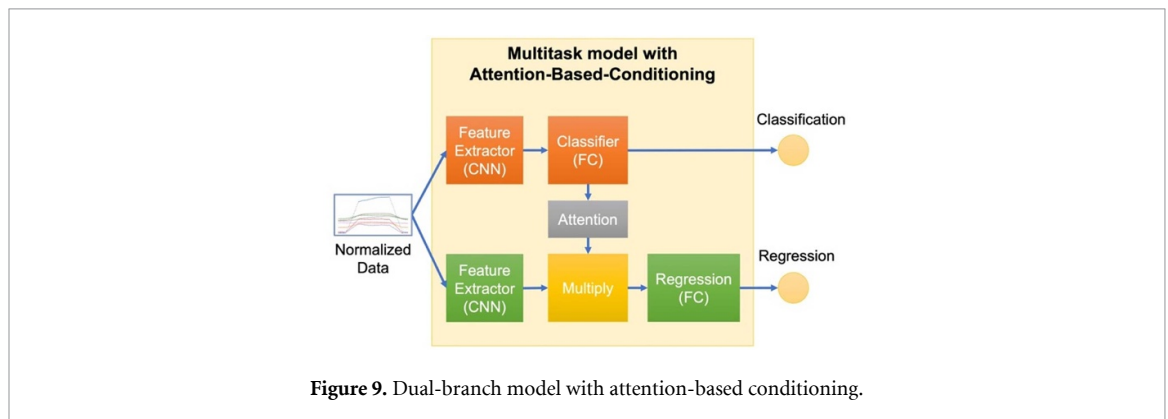


Figure 9. Dual-branch model with attention-based conditioning.

3.1. The proposed dual-branch attention architecture

Our proposed architecture is a dual-branch model with attention-based conditioning, as depicted in figure 9 and table 2. The model takes the normalized sensor time-series as input and processes it through two parallel, specialized feature extractors. The top branch, the Classification Head, is responsible for identifying the analyte from a set of N classes (eight gases plus ambient air). It is implemented as a 1D CNN backbone that processes the input time-series tensor ($115 \text{ time points} \times 8 \text{ sensor channels}$) to extract a latent feature representation. This representation is passed through fully connected (dense) layers to a final softmax activation layer, which produces a probability distribution over the N classes. The bottom branch, the Regression Head, is responsible for estimating the concentration of the gas. It features its own independent 1D CNN backbone, allowing it to learn a feature representation specifically optimized for the nonlinearities of concentration prediction. The selection of 1D CNN backbones instead of architectures like LSTM, TCN, or hybrid CNN-LSTM is driven by their superior ability to autonomously learn hierarchical features from raw thermal fingerprints while preserving full temporal integrity. This choice provides an optimal balance between high-fidelity pattern recognition and the stringent resource constraints of Edge AI systems. Unlike models that require heavy computational overhead to manage complex temporal states—particularly as standard STM32 MCUs currently lack optimized support for efficient LSTM execution—our unified CNN-based design captures discriminative patterns in the shape and evolution of sensor responses while remaining lightweight. This architectural efficiency ensures robust, real-time inference on resource-constrained MCUs without the need for lossy pre-processing, as validated by our successful deployment across 10 different STM32 MCUs.

A more detailed, layer-level schematic of this architecture is provided in figure 10. The model is designed to be lightweight, containing a total of 180 534 trainable parameters with a computational complexity of 1.03 MFLOPS. The Classification Head (top branch) processes the input tensor of shape

Table 2. Architecture of dual-branch model with attention-based conditioning (the bolded numbers indicate the total number of parameters).

Main parts	Blocks	Configurations	Output size	Number of parameters
Input		8 sensor signals	115×8	0
Classification feature extraction	Conv1D-1	Conv1D ($K = 3 \times 3, F = 20$)	115×20	500
		Dropout (0.3)	115×20	0
		BatchNormalization	115×20	80
		ReLU	115×20	0
		MaxPooling ($K = 2, S = 1$)	57×20	0
	Conv1D-2	Conv1D ($K = 3 \times 3, F = 20$)	57×40	2440
		Dropout (0.3)	57×40	0
		BatchNormalization	57×40	160
		ReLU	57×40	0
		MaxPooling ($K = 2, S = 1$)	28×40	0
Classifier	Flatten	Flatten	1120	0
	Dense-1	Dense (32)	32	35 872
		ReLU	32	0
		Dropout (0.3)	32	0
	Classification Output	Dense (9)	9	297
		Softmax	9	0
Attention	Dense-1	Dense (32)	32	320
		ReLU	32	0
	Dense-2	Dense (16)	16	528
		ReLU	16	0
		Dropout (0.3)	16	0
	Dense-3	Dense (896)	896	15 232
Regression feature extraction	Conv1D-1	Conv1D ($K = 3 \times 3, F = 16$)	115×16	400
		ReLU	115×16	0
		MaxPooling ($K = 2, S = 1$)	57×16	0
	Conv1D-2	Conv1D ($K = 3 \times 3, F = 32$)	57×32	1568
		ReLU	57×32	0
		MaxPooling ($K = 2, S = 1$)	28×32	0
Regressor	Flatten	Flatten	896	0
	Dense-1	Dense (128)	128	114 816
		ReLU	128	0
	Dense-2	Dense (64)	64	8,256
		Dropout (0.2)	64	0
	Regression Output	Dense (1)	1	65
				180 534

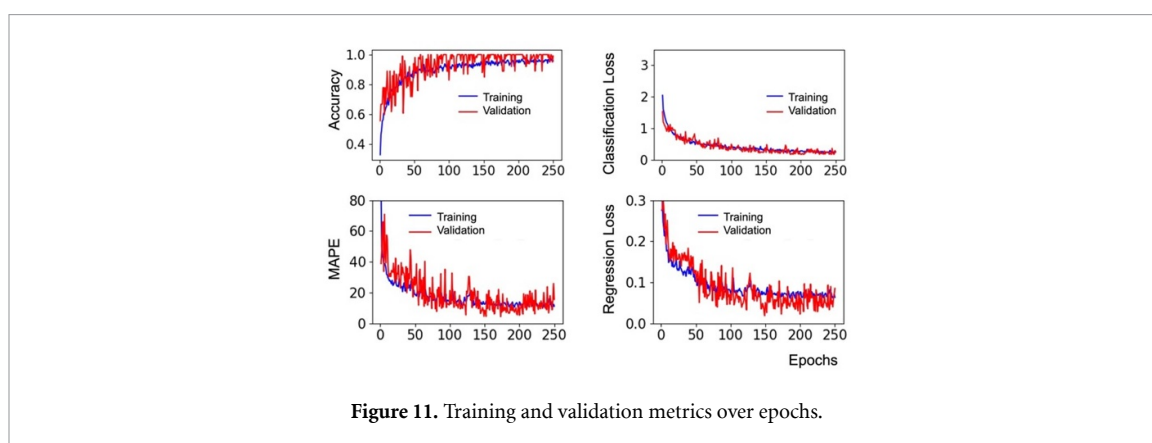
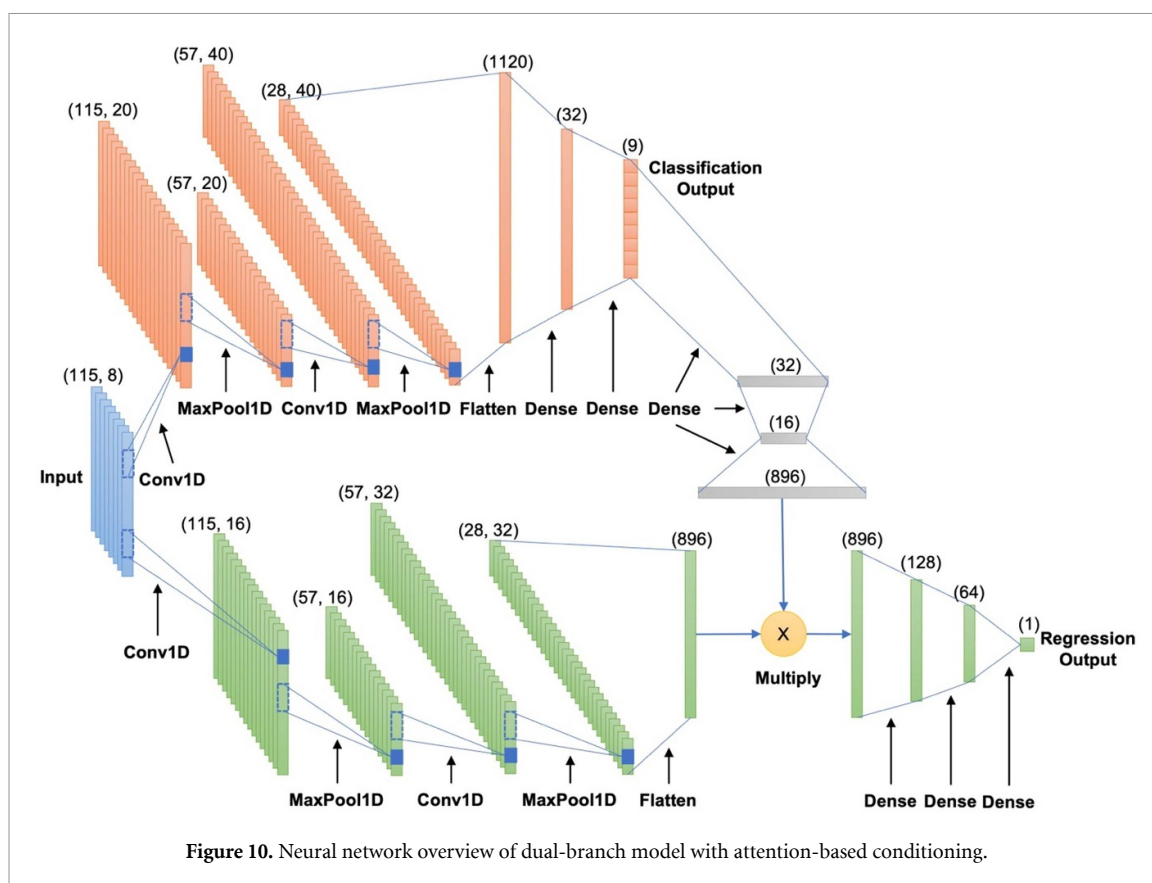
K : Convolutional kernel size.

F : Number of convolutional filters.

S : Strides.

(115, 8) through a sequence of Conv1 and MaxPool1D layers. This reduces the dimensionality while extracting key temporal features, ultimately producing a feature vector of size (896). This vector is then passed through two dense layers, culminating in a 9-unit dense layer with a softmax activation to produce the final class probabilities. The Regression Head (bottom branch) similarly processes the input tensor through its own independent stack of Conv1D and MaxPool1D layers, also producing a feature vector of size (896). The attention mechanism is implemented by taking the softmax output from the Classification Head and passing it through a small multi-layer perceptron (consisting of dense layers of size 32 and 16) to generate a learned attention weight vector of size (896). This attention vector is then applied to the Regression Head's feature vector via element-wise multiplication (Multiply layer). Finally, the modulated feature vector is passed through a terminal MLP (with dense layers of size 256, 128, and 1) to produce the single scalar output representing the estimated gas concentration.

The synergy between these branches is achieved through a novel attention mechanism. This mechanism functionally conditions the regression task on the classification output, allowing the Regression



Head to dynamically adapt its function based on the predicted gas type. The N -dimensional probability vector from the Classification Head is transformed into an attention weight vector, which is then applied via element-wise multiplication to the feature map from the Regression Head's backbone. This operation dynamically re-weights the feature space, selectively amplifying patterns most salient to the predicted analyte.

3.2. Concentration-adaptive loss function

A ubiquitous challenge in chemical sensing is that analyte concentrations often span several orders of magnitude. The standard loss function for regression tasks is the mean absolute error (MAE), which computes the average of the absolute differences between the true (y_{true}) and predicted (y_{pred}) values, as expressed in equation (6):

$$\text{MAE} = \frac{1}{N} \sum |y_{\text{true}} - y_{\text{pred}}|. \quad (6)$$

While simple and effective in many domains, MAE is scale-dependent. In our application, an absolute error of 1 ppm is a minor deviation at a concentration of 100 ppm but a catastrophic failure at a

concentration of 2 ppm. Consequently, a model trained with MAE will be inherently biased, prioritizing the minimization of large absolute errors that occur at high concentrations while neglecting to achieve the high relative accuracy required at low concentrations.

To overcome this fundamental limitation, we engineered a concentration-adaptive loss function. This function dynamically normalizes the error for each sample based on its true concentration magnitude, thereby balancing the learning process across the entire operational range, as expressed in equation (7):

$$\text{Custom Loss} = \frac{1}{N} \sum \frac{|y_{\text{true}} - y_{\text{pred}}|}{(y_{\text{true}} + \varepsilon)^k}. \quad (7)$$

In this formulation, the absolute error for each sample is inversely scaled by its true concentration (y_{true}), modulated by a tunable exponent, k . By setting $k > 0$, the loss function assigns a disproportionately higher penalty to prediction errors on low-concentration samples, forcing the model to achieve high fidelity where it matters most. The term ε is a small constant added for numerical stability to prevent division by zero.

This design is a flexible generalization of the MAPE concept. When $k = 1$ and, it becomes equivalent to MAPE. When $k = 0$, it reduces to the standard MAE loss. By treating k as a hyperparameter, we can systematically optimize the loss function's sensitivity to find the ideal balance for our specific dataset. The performance of the regression task is critically dependent on the hyperparameters k and ε in our custom loss function. To determine their optimal values, we performed an exhaustive grid search. For each combination of candidate values for k and ε , the full MT-DL model was trained across all five cross-validation folds, and the average MAPE was computed on the validation sets. The combination that yielded the lowest average MAPE was selected for the final model evaluation. This data-driven approach ensures that the chosen hyperparameters are optimized for generalization performance.

3.3. Two-stage training protocol

Within each fold of our 5-fold cross-validation scheme, we employed a two-stage training process to optimize the MT-DL model using a batch size of 32 and a maximum of 250 epochs. This two-stage process stabilizes training and allows the regression task to benefit from a high-quality, pre-trained feature representation of the gas identity: 1) Stage 1—Classification Head pre-training: the Classification Head was trained independently on the gas identification task. The model was optimized using the categorical cross-entropy loss function and the Adam optimizer. We employed an adaptive learning rate schedule, which reduces the learning rate upon plateauing of the validation accuracy, and early model checkpointing (using a patience of 20 epochs) to save the parameters corresponding to the best validation performance. This stage ensures the model first becomes an expert at gas identification; 2) Stage 2—full MT-DL model training: after Stage 1, the weights of the trained Classification Head were frozen. The full MT-DL framework, including the Regression Head and the attention mechanism, was then trained. The model was optimized solely on the custom-engineered regression loss function (section 3.2). The same adaptive learning rate and checkpointing strategies were used, this time monitoring the validation regression loss.

3.4. Performance evaluation metrics

To provide a comprehensive assessment of the model, we used a standard set of metrics for both the classification and regression tasks. All reported metrics are aggregated from the predictions made on the held-out test sets from all five folds of the cross-validation. Performance on the gas identification task was evaluated using accuracy, precision, recall, and $F1$ -score, derived from the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) counts, as expressed by equation (8–11). *Accuracy*: the proportion of all predictions that were correct. *Precision*: the proportion of positive predictions that were actually correct. It measures the model's exactness. *Recall (sensitivity)*: the proportion of actual positive samples that were correctly identified. It measures the model's completeness. *$F1$ -score*: The harmonic mean of precision and recall, providing a single, balanced measure of a model's performance. A confusion matrix is also presented for visual inspection of class-wise performance,

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (8)$$

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (9)$$

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (10)$$

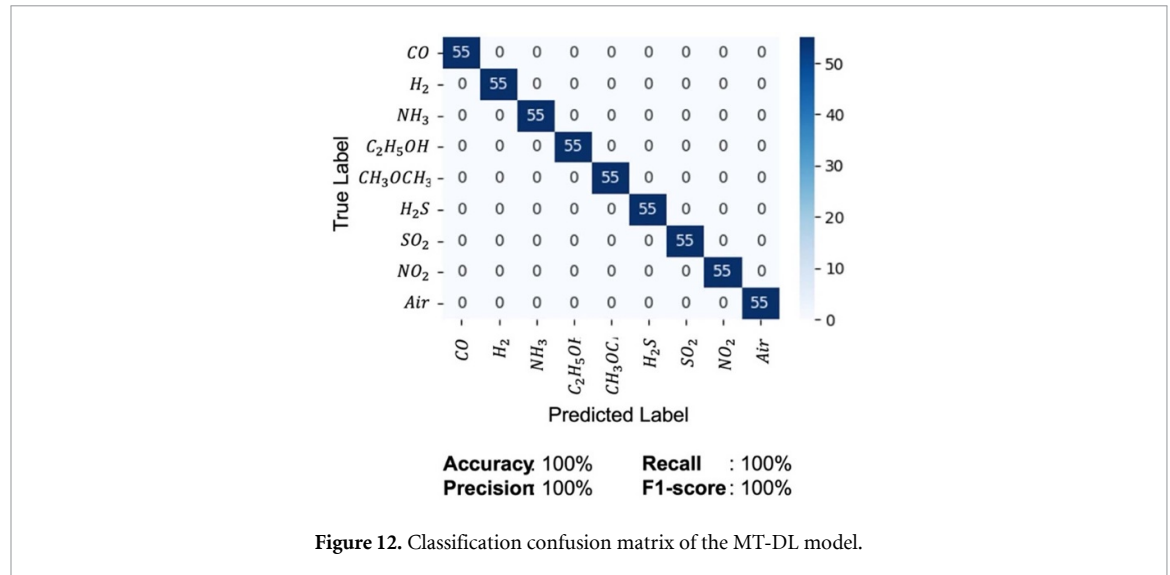


Figure 12. Classification confusion matrix of the MT-DL model.

$$F1 - \text{Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}. \quad (11)$$

Performance on the concentration estimation task was primarily evaluated using MAPE, which provides a scale-independent measure of predictive accuracy. MAPE expresses the average error as a percentage of the true value, as expressed by equation (12). It is particularly useful for evaluating performance across different concentration ranges. The residual standard deviation (RSD) was also computed to assess the magnitude of the prediction errors, as expressed by equations (13) and (12). MAPE per ppm range (MAPE-R) is also used to facilitate a more equitable comparison of performance across gases with different maximum concentration scales (e.g. 0–100 ppm vs 0–10 ppm), as expressed by equation (14). It normalizes a gas's MAPE by its maximum concentration range ($C_{\max}$), providing a measure of relative error per unit of concentration (%/ppm). In addition, we employ visual diagnostics, including predicted vs true concentration plots and histograms of the prediction residuals, to qualitatively assess model fit and bias,

$$\text{MAPE} = \frac{100\%}{N} \times \sum_{i=1}^N \left| \frac{y_{\text{true},i} - y_{\text{pred},i}}{y_{\text{true},i}} \right| \quad (12)$$

$$\text{RSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{\text{true},i} - y_{\text{pred},i})^2} \quad (13)$$

$$\text{MAPE-R} = \frac{\text{MAPE}}{C_{\max}}. \quad (14)$$

4. Results and discussion

4.1. Gas classification performance

Figure 11 illustrates the evolution of the training and validation metrics over epochs for one data split. Both training and validation losses decrease steadily, indicating stable convergence and good generalization. The concurrent improvement in classification accuracy and reduction in regression loss demonstrate effective multitask learning. The classification performance of the MT-DL framework was exceptional. As shown in the confusion matrix in figure 12, the model achieved perfect (100%) accuracy in identifying all eight target gases as well as the 'clean air' class across all test folds. Consequently, the associated metrics of precision, recall, and $F1$ -score were also all 100%. This performance underscores the effectiveness of using the complete temporal fingerprint as input. The deep convolutional network successfully learned highly discriminative features from the dynamic evolution of the sensor responses, a task where classical methods that rely on static, engineered features often falter. The model's ability to robustly distinguish between the unique signatures of each gas provides a solid and reliable foundation for the subsequent regression task.

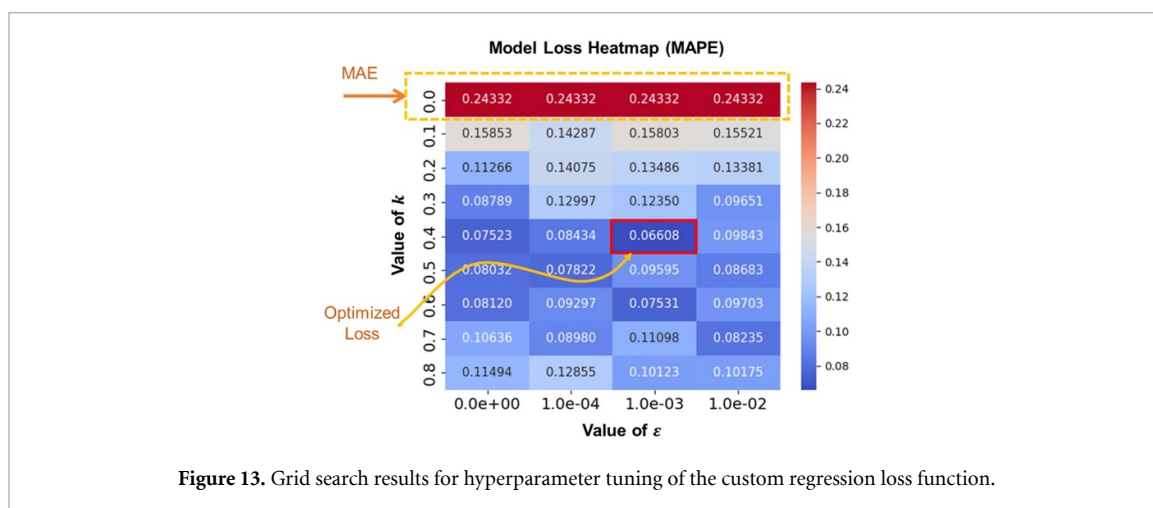


Figure 13. Grid search results for hyperparameter tuning of the custom regression loss function.

Table 3. Regression evaluation results of the MT-DL model (the bolded numbers indicate the average values).

Gas	MAPE (%)	MAPE-R (%/ppm)	RSD (ppm)
CO	9.02	0.09	1.70
H ₂	4.90	0.05	2.73
NH ₃	2.84	0.03	0.81
C ₂ H ₅ OH	5.14	0.05	3.48
CH ₃ OCH ₃	3.02	0.03	1.70
H ₂ S	6.64	0.66	0.14
SO ₂	5.93	0.59	0.18
NO ₂	15.38	1.54	0.48
Average	6.61	0.38	1.40

4.2. Gas concentration regression performance

The regression performance of the complete MT-DL framework was a key success, demonstrating high fidelity across all analytes. A critical innovation of our framework is the custom-engineered loss function, designed to ensure high accuracy across the wide dynamic range of concentrations. To quantify its impact and determine its optimal parameters, we first conducted an exhaustive hyperparameter grid search. The results, visualized in figure 13, clearly show the sensitivity of the model's performance to the loss function's formulation. The optimal configuration ($k = 0.4$, $\varepsilon = 10^{-3}$) yielded the lowest average MAPE. In stark contrast, training the same architecture with a standard MAE loss function (equivalent to setting $k = 0$) resulted in a much higher average MAPE of 24.33%.

The optimization of our custom loss function was the key enabler for the model's final regression performance. With the optimized loss, the complete MT-DL framework achieved an impressive average MAPE of 6.61% across all gases. The detailed per-gas results, presented in table 3, provide a nuanced view of the model's performance when analyzed through the complementary metrics of MAPE, MAPE-R, and RSD.

The model demonstrated outstanding performance for several analytes. For example, NH₃ achieved a low MAPE of 2.84%, a very low MAPE-R of 0.03%/ppm, and a sub-ppm RSD of 0.81 ppm, indicating high accuracy in both relative and absolute terms. A more insightful analysis emerges when comparing gases across different concentration scales. For CO, measured over a wide 100 ppm range, the MAPE was a moderate 9.02%. However, its MAPE-R was only 0.09%/ppm, indicating the error was small relative to the large scale. The RSD of 1.70 ppm provides the absolute context for this error magnitude. Conversely, the model's absolute precision is best highlighted on the narrow-range gases. For H₂S and SO₂, the model achieved extremely low RSDs of only 0.14 ppm and 0.18 ppm, respectively. This demonstrates exceptional precision in absolute terms. While their MAPEs (6.64% and 5.93%) appear moderate, this is a mathematical consequence of the small 10 ppm denominator; the low RSD values confirm that the model's predictions are highly accurate. The metrics also clearly identify NO₂ as the most challenging analyte. It exhibited the highest MAPE (15.38%), the highest MAPE-R (1.54%/ppm), and a comparatively high RSD (0.48 ppm relative to its 10 ppm range). This suggests its sensor response may contain more complex nonlinearities that could benefit from further targeted investigation.

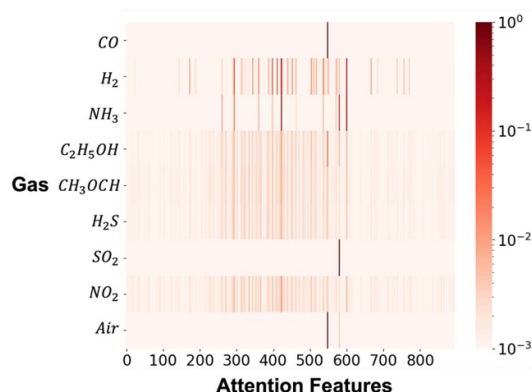


Figure 14. Visualization of the gas-specific attention weight vectors across the 896-dimensional feature space.

The detail quantitative performance is qualitatively shown in figure 15, which shows the predicted versus true concentration plots and residual histograms for each gas. The predictions cluster tightly around the ideal diagonal line with residuals centered near zero, indicating that the model is both accurate and unbiased.

4.3. Interpretation of the attention mechanism

To understand the MT-DL framework's efficacy, we analyzed the 896-dimensional attention vectors that modulate regression features via element-wise multiplication. Figure 14 visualizes these gas-specific weight vectors across the feature space. For CO, Air, and SO₂, the mechanism is highly selective, focusing on sharp temporal patterns with peak weights reaching 1.00. This high sparsity indicates that the model identifies unique fingerprints for these analytes to optimize quantification, resulting in exceptional precision, such as an RSD of 0.18 ppm for SO₂. Conversely, for gases like H₂ and NH₃, weights are more distributed across the feature space (peaks ~0.36–0.41), indicating the integration of broader sensor interaction patterns to resolve complex signatures. This dynamic reweighting confirms that the classification branch effectively guides the regression task, allowing the model to adapt its feature focus for each specific analyte.

4.4. Ablation study: impact of model architecture

To validate our choice of the dual-branch attention architecture, we conducted an ablation study comparing its performance against two logical alternative multitask designs. The first alternative, a shared feature extractor model (figure 16), yielded a poor MAPE of 23.98%. This design creates a representational bottleneck, forcing a single CNN backbone to learn a compromised set of features that are not optimal for either classification or regression, leading to significant performance degradation. The second alternative, a dual feature extractor with feature concatenation (figure 17), performed better with a MAPE of 21.08%. By allowing for specialized feature learning in separate backbones, it overcame the bottleneck issue. However, the static concatenation of features still limits the model's ability to adapt. Our proposed dual-branch attention architecture substantially outperformed both alternatives with a final MAPE of 6.61%. This demonstrates that the dynamic conditioning provided by the attention mechanism is the critical component for success. It allows the model to not only learn specialized features but also to synergistically combine them in a targeted, adaptive manner based on the identified gas. The results, summarized in table 4, definitively establish the superiority of our proposed approach.

4.5. Comparison with classical ML benchmarks

To benchmark our MT-DL framework, we compared its performance against the classical ML models previously studied for this task. As detailed in

Table 5, the best-performing classical model (RF without PCA) achieved a maximum accuracy of 83.84%. Our MT-DL model's 100% accuracy represents a significant leap in performance, highlighting the limitations of classical algorithms in processing high-dimensional, temporal data. For concentration estimation, a baseline linear regression (LR) model yielded a prohibitively high average MAPE of 69.71% (table 6). The 6.61% MAPE achieved by our MT-DL model underscores the highly nonlinear nature of the problem and the necessity of a deep learning approach to effectively model it.

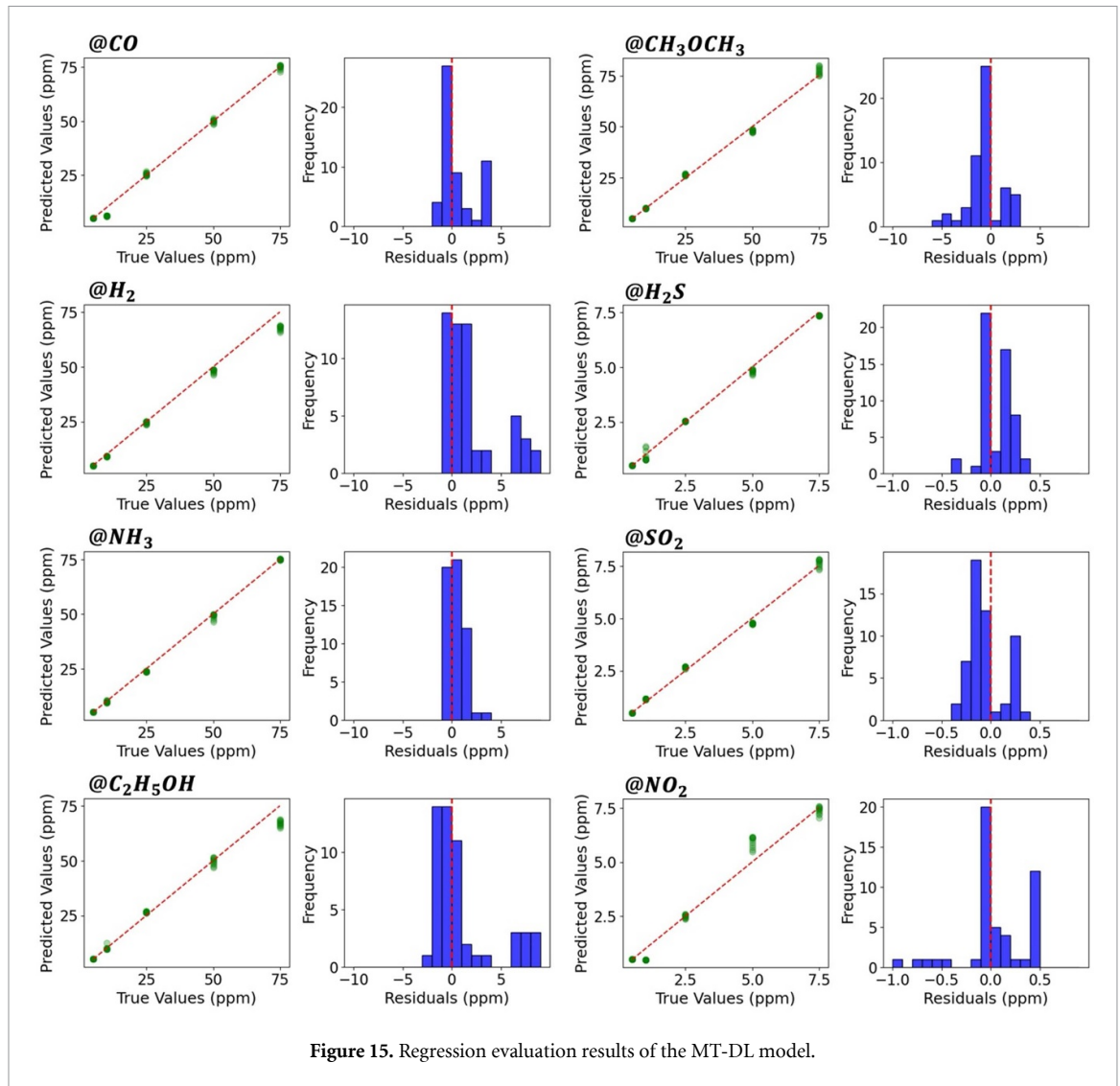


Figure 15. Regression evaluation results of the MT-DL model.

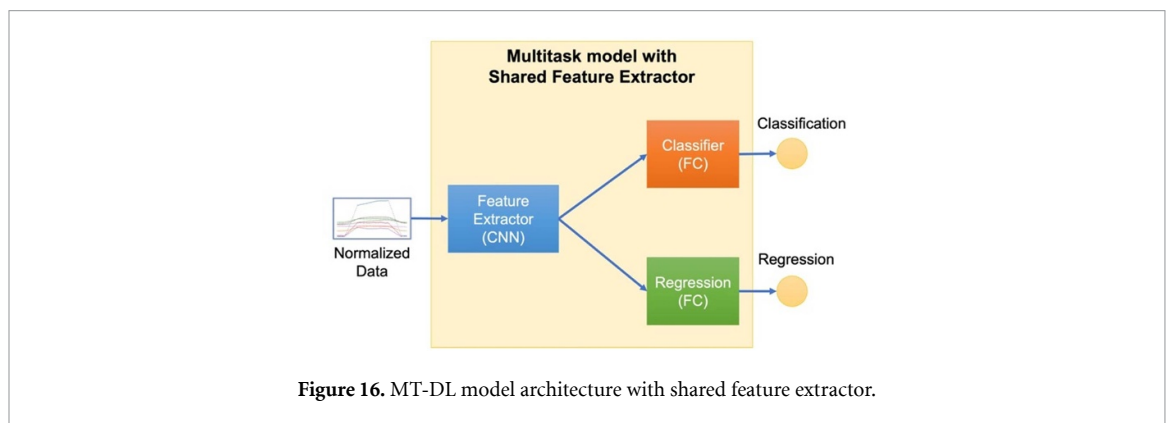
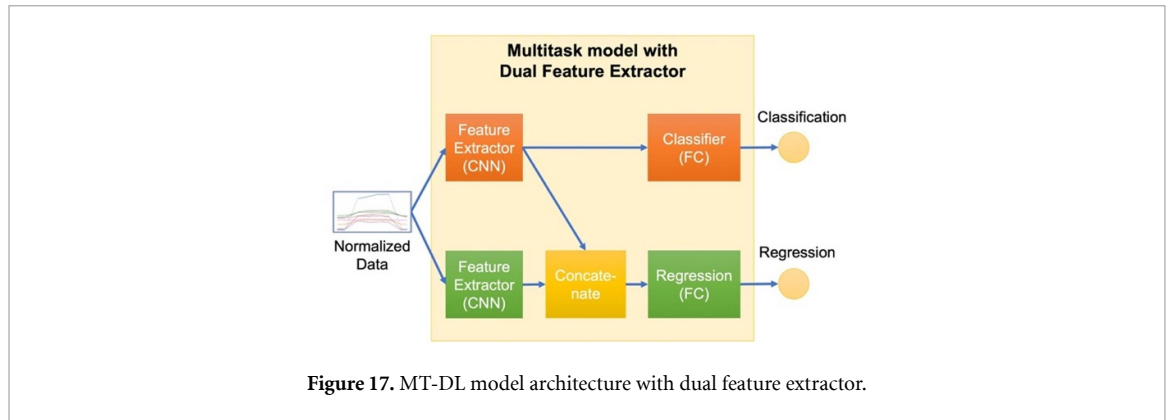


Figure 16. MT-DL model architecture with shared feature extractor.

4.6. Deployment feasibility on embedded systems

A critical aspect of our framework's evaluation is its practical viability for real-world e-nose systems, which are typically deployed on resource-constrained embedded platforms. The model has only 180 534 parameters, 1.03 MFLOPS as the computational cost, and 0.514 MMACS as the multiply-accumulate count. To this end, we benchmarked the MT-DL model's inference performance across a diverse suite of 10 STM32 MCUs using the ST Edge AI Developer Cloud, a specialized environment for optimizing and benchmarking AI models on STM32 MCUs. This evaluation, inspired by the methodology in, covers a wide spectrum of computational capabilities, with maximum operating frequencies ranging from a

**Table 4.** Performance comparison of MT-DL architectures (the bolded numbers indicate the best performance).

Architecture	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	MAPE (%)
Shared feature	73.13	69.82	73.13	69.89	23.98
Dual feature	75.76	73.35	75.76	72.51	21.08
Attention feature	100	100	100	100	6.61

Table 5. Classification evaluation results: comparison between baseline ML models and our MT-DL model (the bolded numbers indicate the best performance).

Models	Without PCA				With PCA			
	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
kNN	65.86	72.39	65.86	66.29	59.80	63.34	59.80	60.06
LSVM	81.01	83.77	81.01	81.50	80.61	83.85	80.61	81.02
KSVM	60.20	71.91	60.20	62.09	58.38	70.39	58.38	59.74
LDA	58.99	66.58	58.99	59.99	66.67	75.21	66.67	67.57
MLP	75.56	79.11	75.56	76.13	76.97	80.68	76.97	77.13
RF	83.84	87.48	83.84	83.90	65.05	70.08	65.05	66.38
DT	76.97	78.57	76.97	77.06	65.25	67.30	65.25	65.47
GNB	76.35	78.89	76.35	76.11	73.54	77.14	73.54	73.61
QDA	72.73	80.73	72.73	73.64	77.78	82.84	77.78	78.51
GB	73.54	76.93	73.54	73.34	64.85	70.41	64.85	66.35
MT-DL Model (Ours)	100	100	100	100	Do not need to apply PCA because it takes time for preprocessing data			

Table 6. To benchmark our MT-DL framework, we compared its performance against the classical ML models previously studied for this task (the bolded numbers indicate the best performance).

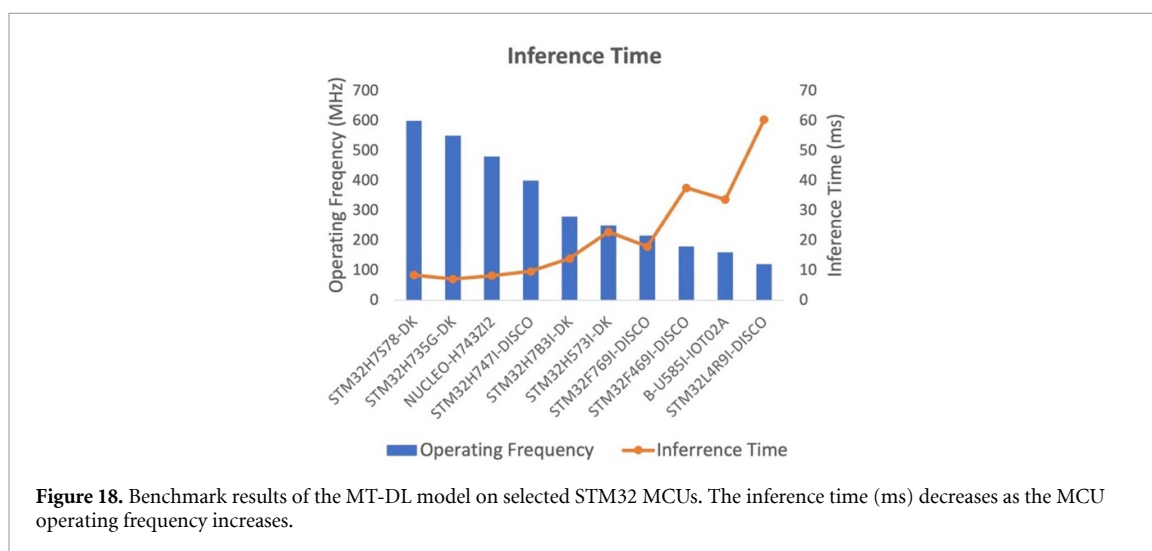
Gas	Linear regression ([33]) (%)	MT-DL model (ours) (%)
CO	115.69	9.02
H ₂	27.65	4.90
NH ₃	17.25	2.84
C ₂ H ₅ OH	10.53	5.14
CH ₃ OCH ₃	32.00	3.02
H ₂ S	95.35	6.64
SO ₂	85.24	5.93
NO ₂	173.95	15.38
Average	69.71	6.61

low-power 120 MHz to a high-performance 600 MHz. To preserve maximum fidelity and provide a conservative estimate of performance, the model was deployed without any form of quantization, pruning, or weight compression.

The results (in table 7) demonstrate that the model achieves real-time performance across this entire spectrum. On a high-performance 550 MHz STM32H735G, the model achieved a swift inference time of

Table 7. Performance metrics of the MT-DL model across 10 STM32 microcontrollers (the bolded numbers indicate the best and worst performances).

MCU	Flash memory usage (KB)	RAM usage (KB)	Inference time (ms)	Throughput (predictions/sec)
STM32H7S78-DK	714.12	18.75	8.43	118
STM32H735G-DK	714.12	18.75	7.11	140
NUCLEO-H743ZI2	714.12	18.75	8.21	121
STM32H747I-DISCO	714.12	18.75	9.64	103
STM32H7B3I-DK	714.12	18.75	13.99	71
STM32H573I-DK	714.12	18.81	22.84	43
STM32F769I-DISCO	714.11	18.75	17.92	55
STM32F469I-DISCO	714.22	18.75	37.55	26
B-U585I-IOT02A	714.22	18.81	33.63	29
STM32L4R9I-DISCO	714.22	18.75	60.35	16



only 7.11 ms. Crucially, even on a resource-constrained 120 MHz STM32L4R9I, the latency remained a practical 60.35 ms, well within the requirements for continuous monitoring applications and highly sufficient for real-time operation in gas sensing applications. The relationship between computational power and performance is clearly visualized in figure 18, which shows a strong inverse correlation between MCU operating frequency and inference time. This predictable scalability confirms the model's efficient design.

This successful deployment provides the definitive validation that the high accuracy of our MT-DL framework was not achieved at the cost of prohibitive computational complexity. It confirms that our design is not merely an academic exercise but a viable and efficient blueprint for the next generation of intelligent, deployable chemical sensors.

5. Conclusion

This work introduced and validated a unified, end-to-end MT-DL framework, presenting a definitive solution to the critical limitations of conventional ML in e-nose systems. By moving beyond reductive feature engineering and disjointed modeling, our approach demonstrated exceptional performance, achieving 100% classification accuracy and an average regression MAPE of 6.61%. The success of this framework is built upon three synergistic innovations. First, by processing the complete temporal fingerprints, our model avoids the lossy pre-processing inherent in classical approaches. Second, our novel attention-based multitask architecture enabled the synergistic learning of classification and regression, proving to be substantially more effective than alternative designs. Third, the custom-engineered loss function proved essential for achieving high-fidelity quantification across the entire dynamic range, a critical challenge in chemical sensing. The culmination of these design choices is a model that is not only highly accurate but also, as demonstrated by its successful deployment on 10 STM32 MCUs, computationally efficient enough for practical, real-time embedded applications.

Collectively, these advancements establish a new performance benchmark for AI-driven chemical sensing. Our work provides a validated blueprint for creating intelligent sensors that are both highly accurate and viable for deployment at the network edge. While this study was conducted on single-analyte gases under controlled conditions, the robust and scalable methodology presented here provides a strong foundation for future research. The most critical next step is to extend this framework to the significantly more complex challenge of identifying and quantifying constituents within mixed-gas environments, a key requirement for real-world environmental and industrial monitoring.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://github.com/quenx-rd/gas_identifier_public [47].

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ORCID iDs

Xuanque Nguyen  0009-0001-2635-6553
Ngoc Viet Nguyen  0000-0002-1710-9331
Chien Nguyen Viet  0000-0002-1625-5513
Van Su Luong  0000-0003-2061-7406
Quang Vuong Pham  0000-0002-3819-6016
Viet Thong Le  0000-0002-4636-5094
Van Hieu Nguyen  0000-0002-9613-9108
Minhhuy Le  0000-0001-6152-6215

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